

Adaptive Multi-Order Penalized Splines

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Abstract

Regression splines are flexible linear models used for estimating unknown regression functions from noisy data. When the number and location of the knots are optimized, the approximation error is substantially decreased. However, finding good knot locations can be a difficult problem, especially when the unknown regression function is spatially heterogeneous. We present a new approach for adaptively locating knots for “sharp” regression features. These features are identified with zero-order knots which permit jump discontinuities in the regression spline model. The zero-order knots augment a smoother, linear regression spline, which, together, we call the adaptive multi-order penalized spline (AMPS) procedure. The AMPS estimator is shown to perform well compared to competing methods in several simulations. Additionally, we construct a robust estimator of the unknown variance. Like AMPS, this method utilizes a zero-order basis to identify and estimate heterogeneous features.

KEY WORDS: Basis Economy; False Detection Rate; Linear model; Spatially Adaptive; Threshold Selection; Variance Estimation;

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1 Introduction

Regression splines are popular statistical methods because of their excellent approximation properties and simplicity (de Boor 1978). However, they have limited ability to adapt to data characterized by strong varying curvature or smoothness. In this paper we study a multiresolution-type extension of traditional penalized spline that we call the adaptive multi-order penalized splines (AMPS). What is new is the multi-order regression spline design that sequentially adapts to features of increasing smoothness. The resulting spatial adaptivity of AMPS produces low risk fits for a wide variety of regression functions.

Suppose we observe n noisy samples from a function $Y_i = m(x_i) + \epsilon_i$, where x_i is a fixed, univariate covariate and $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ for $i = 1, \dots, n$, and variance σ^2 . The p -th order regression spline is

$$f(x|\beta) = \beta_0 + \beta_1 x + \dots + \beta_p x^p + \sum_{j=1}^k \beta_{j+p} (x - \kappa_j)_+^p, \quad (1)$$

where $p \geq 0$ is an integer, $(x)_+^p = x^p I(x \geq 0)$, and $\kappa_1 \leq \dots \leq \kappa_k$ are the knots. This spline has $p - 1$ continuous derivatives. When fitting this model to data, the key specification is the number and location of the knots.

Recent spline research has focused on two distinct strategies to adapt to regions of spatial heterogeneity. One approach adapts by choosing more basis functions in data-dense regions where the underlying true function has more structure. Successful approaches include the adaptive regression spline methods of Friedman and Silverman (1989) [TURBO], Friedman (1991) [MARS], and Stone, Hansen, Kooperberg, and Troung (1995). Bayesian approaches consider the number and location of knots as unknown parameters to be estimated (Denison, Mallick and Smith 1998 and Hansen and Kooperberg 2002). The other approach solves the knot selection problem by fitting a saturated model and then shrinking knot coefficients towards zero with a penalized fitting criterion. Penalized methods include the cubic smoothing splines of Reinch (1967), Wahba (1990) and the

penalized regression spline formulations of Eiler and Marx (1996) and Ruppert and Carroll (1997). This solution is often simpler and computationally more efficient than stepwise knot addition and deletion procedures.

AMPS is a hybrid procedure that combines features of both appealing adaptive strategies. Like other knot-selection techniques, AMPS adapts to spatial heterogeneity by adding knots in those regions of the data that exhibit the greatest curvature. However, unlike other methods, our knot selection focuses on locating and modeling only very sharp regression features. When sharp features are identified, they are modeled with similarly sharp basis functions, $(x - \kappa)_+^0$, the so-called zero-order knots. First differences of the data are used to efficiently estimate potential jumps in the regression function. The locations of the zero-order knots are then determined by testing for significant jumps while controlling the false detection rate (FDR) (Benjamini and Hochberg 1995 and Storey 2001). This focused knot selection scheme has two main advantages. First, this testing problem locates *all* zero-order knots in a single step and does not require the model to be refit. So we are able to avoid the type of iterative model evaluations that characterize adaptive frequentist methods as well as the iterative MCMC or reversible jump MCMC calculations performed by Bayesian methods. Second, we are able to explicitly control the rate at which knots are falsely selected, i.e. the false detection rate. Theoretical results show that this first stage of AMPS is itself asymptotically minimax (see Section 3.2.)

The second step of AMPS involves adding the zero-order knots to a regression spline with a relatively coarse grid of first degree knots. This step serves to refine the regression spline basis by allowing for additional smoothness between any zero-order features. The resulting multi-order design is then fit with penalized least squares. As such, AMPS can be thought of as an extension to the penalized regression splines (PRS) of Eilers and Marx (1996) and Ruppert and Carroll (1997) that expands the class of well estimated regression functions to include regions of severe spatial heterogeneity.

Conditioned on a set of knots and smoothing parameter(s), regression splines may be written as a linear models (Buja, Hastie and Tibshirani 1989 and Beran 2000). From the linear model perspective it is easy to extend AMPS. This includes the addition of penalized and unpenalized covariates, application within more complicated linear designs, and extensions to nonnormal responses. Additionally, AMPS's second stage penalized multi-ordered regression spline can be formulated as a linear mixed models (Ruppert et al. 2003). Within this framework smoothing parameters are ratios of variance components that can be estimated by ML or REML using standard software tools.

In most regression problems the estimation of the conditional mean and the residual variance are inextricably linked. In the context of smoothing spatially heterogeneous regression functions, this means that a failure to identify sharp features will bias *both* the conditional mean and the variance estimates. To remedy this, we again use the first differences to exclude those residuals that are most contaminated by sharp feature signal. By calibrating the variance estimate with a modest amount of extra variation we are able to automatically determine which residuals to exclude. The resulting variance calibration (VC) variance estimator is a robust, general purpose variance estimator. In the context of the AMPS methodology, the VC estimator plays an important role in the FDR thresholding of the first differences and the computation of the estimated risk used to select the smoothing parameter, both of which depend on a good estimate of variability.

The rest of the article is organized as follows. In Section 2 we review PRS and in Section 3 introduce AMPS as an extension of PRS and provide theoretical support for its good performance. In order to focus attention on the estimation of the conditional mean, we assume that the variance is known in Sections 2 and 3. In Section 4 we explain the VC estimation of σ^2 . Section 5 presents the results from several simulation tests designed to highlight the performance of AMPS with VC variance estimation and compare this to other methods in the literature. In the final section we discuss extensions to the method.

2 Penalized Regression Splines

Suppose we observe n noisy samples from a function $Y_i = m(x_i) + \epsilon_i$, where x_i is a fixed, univariate covariate and $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ for $i = 1, \dots, n$, and variance σ^2 . For notational convenience, let $\theta = (m(x_1), \dots, m(x_n))'$, and write $Y \sim N_n(\theta, \sigma^2 I)$. The goal of this homoscedastic multivariate normal mean problem is to estimate the true but unknown regression function, θ , with small squared error when σ^2 is assumed known. And for the duration of this section we will assume that the regression function is spatially homogeneous. In Section 3 we will relax the assumption of spatial heterogeneity and in Section 4 relax the assumption of known variance.

Following Ruppert and Carroll (1997) the p -th order penalized regression spline is

$$f(x|\beta) = \beta_0 + \beta_1 x + \dots + \beta_p x^p + \sum_{j=1}^k \beta_{j+p} (x - \kappa_j)_+^p, \quad (2)$$

where $p \geq 0$ is an integer, $(x)_+^p = x^p I(x \geq 0)$, and $\kappa_1 \leq \dots \leq \kappa_k$ are the knots. This spline has $p - 1$ continuous derivatives.

Define the design matrix, X , to be $n \times (p + k + 1)$ with i th row

$$X_i = [1 \quad x_i \quad \dots \quad x_i^p \quad (x_i - \kappa_1)_+^p \quad \dots \quad (x_i - \kappa_k)_+^p].$$

The k knots are located at equally spaced sample quantiles. The penalty matrix, D , is any $n \times (p + k + 1)$ matrix. The penalized least squares (PLS) estimate of β is

$$\hat{\beta} = \min_{\beta} (|Y - X\beta|^2 + \gamma |D\beta|^2) = (X'X + \gamma D'D)^{-1} X'y \quad (3)$$

for a given smoothing parameter, γ , where $|\cdot|$ is the quadratic norm. The corresponding PLS estimator of θ is $\hat{\theta} = X\hat{\beta}$, that is,

$$\hat{\theta} = \hat{m}(x) = X(X'X + \gamma D'D)^{-1} X'y. \quad (4)$$

Because the smoothing parameter γ controls the amount of smoothing, the value of k is not crucial as long as there are enough knots to capture the structure of the regression function. Ruppert (2002) suggests using between 5 and 40 knots depending on the complexity of the regression function.

2.1 Canonical Form

Writing the PRS estimator in canonical form facilitates computation and comparison with other estimators. In canonical form, PRS's can be considered linear smoothers (Buja, Hastie and Tibshirani 1989), a REACT estimators (Beran 2000, 2001), or generalized ridge estimators (Hoerl and Kennard 1970). The following sequential decomposition reduces any of these estimators to canonical form.

Let R be the Cholesky factor of $X'X$, i.e. $R'R = X'X$. Then

$$\hat{\theta} = XR^{-T}(I + \gamma R^{-T}D'DR^{-1})^{-1}R^{-1}X'y. \quad (5)$$

Let $\Gamma\Lambda\Gamma'$ be the spectral decomposition of the symmetric matrix $R^{-T}D'DR^{-1}$ where Γ is $(p+k+1) \times (p+k+1)$ orthonormal and $\Lambda = \text{diag}(\lambda_i)$ is the diagonal matrix with the eigenvalues ordered $0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{p+k+1}$. Then it follows that

$$\hat{\theta} = XR^{-T}\Gamma'(I + \gamma\Lambda)^{-1}\Gamma R^{-1}X'y. \quad (6)$$

Let $U = XR^{-T}\Gamma'$ be the smoothing basis. Define the shrinkage vector to be $f(\gamma) = \text{diag}(1/(1 + \gamma\lambda_1), \dots, 1/(1 + \gamma\lambda_p))$. The canonical form of the estimator is

$$\hat{\theta} = U\text{diag}(f(\gamma))U'y.$$

If $z = U'y$, then $z \sim N_{p+k+1}(\eta, \sigma^2 I)$ where the vector of canonical means $\eta = U'\theta = E(U'y)$. The canonical estimator can be written $\hat{z} = U'\hat{\theta} = \text{diag}(f(\gamma))z$.

In this form the linear estimator has two distinct components, the orthogonal smoothing basis, U , and the shrinkage vector, f . This section concludes with a brief discussion of the components and the role they play in adaptive smoothing.

2.1.1 The Smoothing Basis

The smoothing basis is itself composed of the design matrix, X , and the penalty matrix D . The smoothing basis has the same column space as the design, X , defined here by the p th-order re-

gression spline with knots κ . This basis is a convenient tool with which we can control both the location and smoothness of regression features to be modeled.

The penalty matrix defines lower dimension model commonly called the shrinkage target. We use the penalty matrix equal to a diagonal matrix with $p + 1$ zero's corresponding to first $p + 1$ polynomial regression bases followed by k ones $D = \text{diag}(0, \dots, 0 \quad 1, \dots, 1)$. This places a penalty on the p -th derivatives of the regression curve at the knots. The implied shrinkage target is a p degree polynomial.

Other penalty matrices have been suggested in the literature. Beran (2001) uses difference operators for equally spaced data and local polynomial annihilators for unequally spaced data. Heckman and Ramsay (2000) investigate a wide class of linear differential operators for penalties in the context of smoothing splines. The L_q penalties of Ruppert and Carroll (2000) define another class of diagonal matrices.

2.1.2 The Shrinkage Vector and the Smoothing Parameter

The shrinkage vector f defines the path that connects the unpenalized, least-squares model with the penalized shrinkage target model. In the canonical PLS model, this path is parameterized by the smoothing parameter γ which controls the quality and degree of the shrinkage.

The choice of smoothing parameter, γ , is an important determinant in producing low risk fits to the data. The PRS adapts over the shrinkage vector by using a data based method to select γ . In this paper we select the smoothing parameter by minimizing the estimated risk. Other choices for smoothing parameter estimation include GCV and Mallow's Cp.

The risk of the PRS under squared error loss is $E \sum_{i=1}^n (\theta_i - \hat{\theta}_i)^2$. Since the squared error loss is invariant under orthogonal transformations the risk in canonical form is

$$R(\gamma) = E \left[\sum_{i=1}^n (\eta_i - f_i(\gamma) z_i)^2 \right] = \sum_{i=1}^n ((1 - f_i(\gamma))^2 \eta_i^2 + f_i(\gamma)^2 \sigma^2).$$

Assuming that the variance σ^2 is known, the risk depends on unknown quantity, η_i^2 . We estimate

η_i^2 with $(z_i^2 - \sigma^2)$ to compute the estimated risk as

$$\hat{R}(\gamma) = \sum_{i=1}^n ((1 - f_i(\gamma))^2 (z_i^2 - \sigma^2) + f_i(\gamma)^2 \sigma^2). \quad (7)$$

Letting $g_i = (z_i^2 - \sigma^2)/z_i^2$ the estimated risk can be simplified

$$\hat{R}(\gamma) = \sum_{i=1}^n \left[f_i(\gamma) - g_i \right]^2 z_i^2 + \sigma^2 \sum_{i=1}^n g_i.$$

Thus, the estimated risk minimization problem can be solved with standard software as a weighted least squares minimization of $\sum_{i=1}^n (f_i(\gamma) - g_i)^2 z_i^2$ (Beran 2001).

Many other approaches to shrinkage and thresholding with linear models have been proposed. See Hastie et al. (2001) for an introduction to shrinkage methods. For connections to the the linear mixed model see Ruppert et al. (2003).

2.1.3 Basis Economy and Adaptive Regression

Beran (2000) introduced the idea of basis economy (BE) to describe the ability of a particular basis to fit noisy data. The orthogonal smoothing basis, U , is said to be economical if all but the first few components of η are zero (Beran 2000). The most economical basis is achieved when the dominant eigenvector is proportional to the unknown mean, θ . Thus, basis economy results in a sparse vector η and efficient data compression. Regression estimation is simplified when the basis is economical. Not surprisingly, basis economy plays an important role in the characterization of asymptotic performance of the linear estimators. Beran (2000) proves that lower bound on the minimax risk of a linear estimator is a function of the basis economy for a given estimator.

If most of the signal in η is described by relatively few components, the signal may be effectively estimated by identifying and estimating only these large components, producing low-risk fits to the data. The estimates of the remaining components are fixed at, or near, zero in order to eliminate the noise contaminating the original regression function. This estimation strategy adds to the overall risk some relatively small biases from ignoring the nearly zero components of η , but it also

eliminates the many variances that would arise from an attempt to estimate these unbiasedly, as least squares does (Beran 2000). This is the basic adaptive strategy.

BE plots are index plots of the ordered absolute values of the canonical response $|z_i|$. These plots are a good way to visualize the relative economy and strength of a signal detected with a given basis. The largest components of the BE plot represent the signal, with respect the given basis U . Small components represent the noise in the data. In the middle are those components that the basis is can not separate into just signal or noise. Thus, maximum and minimum economy will produce plots of a single spike or a uniform distribution, respectively.

While Beran (2000) used basis economy as a tool for comparing the utility of different bases for a given problem, we use basis economy to identify and isolate particular features of the noisy data set. These uses of BE are central to the development of both the AMPS and VC estimators.

3 Adaptive Multi-Order Penalized Splines

The adaptive multi-order regression spline (AMPS) method is an extension of the traditional constant-order PRS designed to adapt to regions of strong spatial heterogeneity. Thus, AMPS inherits many attractive properties from its PRS progenitor, including computational efficiency, easy implementation, and good performance. In what follows, we describe the multiorder regression spline model and the algorithm used to build and fit the model.

Linear and quadratic PRS are good models for approximating smooth functions precisely for the same reason that they are inadequate for fitting sharp regression features: they rely on a relatively coarse set of smooth knots. In order to better model sharp features we consider the class of multi-order regression splines,

$$f(x|\beta, \alpha) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p + \sum_{j=1}^k \beta_{j+p} (x - \kappa_j)_+^{\alpha(j)} \quad (8)$$

where the smoothness of the j th knot is indexed by $\alpha(j)$ and p is a nonnegative integer. This spline

is more flexible than (1) because it may take on jumps in the $\alpha(j)$ derivative at the j -th knot. Thus, p -th order regression splines are specified with $\alpha(j) = p$ for all $j = 1, \dots, k$. Once p, κ, α have been specified, the multiorder spline is fit with penalized least squares.

Although $\alpha(j)$ can, in theory, be any nonnegative value, it is common practice to consider only smoothness orders $\alpha(j) = 0, 1$, or 2 . The reason for this limited set of smoothness orders is that they are sufficient for describing most local regression features: zero-order knots model discontinuous jumps, linear knots control the local trend through the first derivative, and quadratic knots control the local curvature through the second derivative. Our approach is to augment a smooth linear or quadratic regression spline basis with ‘zero-order’ knots capable of fitting sharp regression features such as jumps and spikes.

3.1 AMPS Algorithm

The AMPS algorithm is an ordered, two-step process. The first step adaptively identifies zero-order structure in the data. This is an exploratory step that serves to control bias introduced when the smoothing basis is not flexible enough to fit sharp local features. In the second step the multiorder regression spline is fit with penalized least squares, adapting over the shrinkage vector. This can step can be thought of as the estimation step of the algorithm, penalizing most those bases least supported by the data.

1. Basis Adaptation

Calculate first differences $d_i = y_{i+1} - y_i$, $i = 1, \dots, n - 1$. Locate zero-order knots where non-zero first differences occur. Test for non-zero first differences while controlling the false detection rate. Let $\omega_1, \dots, \omega_m$ be the selected zero-order knots for nonnegative integer m .

2. Shrinkage Vector Adaptation

Add the zero-order knots $\omega_1, \dots, \omega_m$ to a linear regression spline and fit with penalized least

squares. The i th row of the multi-order regression spline design matrix is

$$X_i = [1 \ x_i \ I(x_i > \omega_1) \cdots I(x_i > \omega_m) \ (x_i - \kappa_1)_+^1 \cdots (x_i - \kappa_k)_+^1].$$

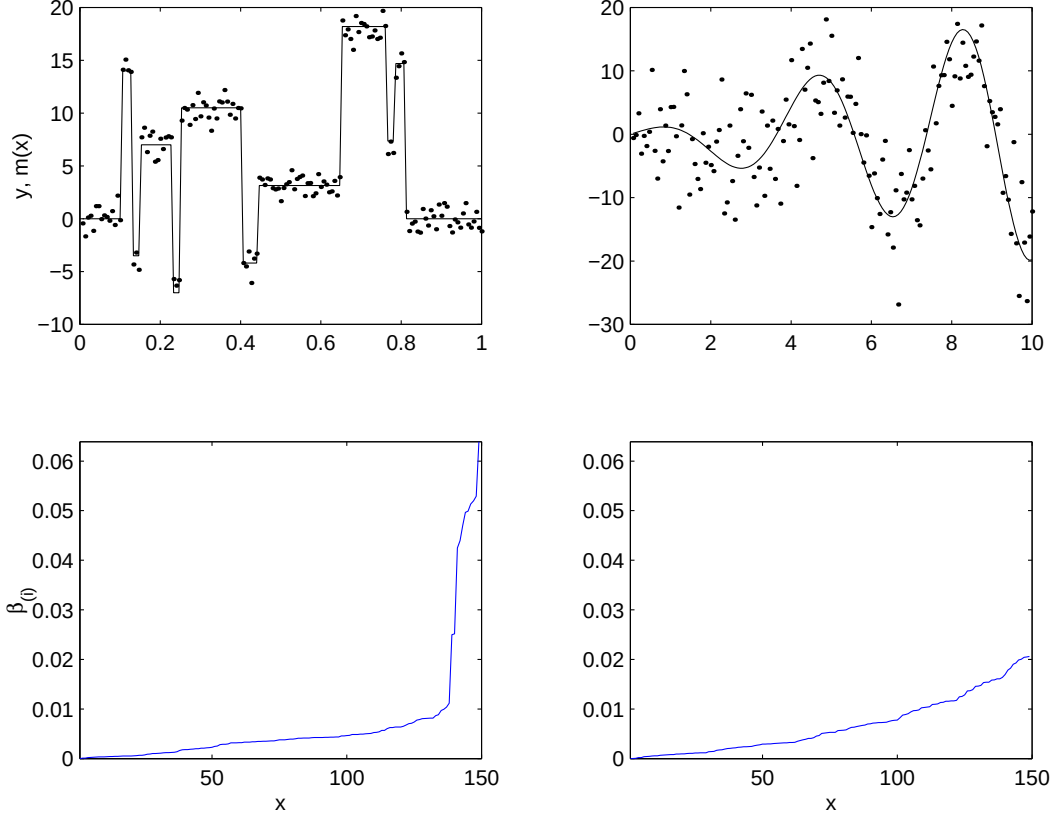


Figure 1: Basis economy plots for piecewise constant and sinusoidal functions. The true function with noisy data (top) and Economy plots $|d_{(i)}|$ (below).

The saturated zero-order regression spline basis

$$X_i = [I(x_i - x_1) \cdots I(x_i - x_n)], \quad (9)$$

with $k = n$ knots, is a good design for identifying sharp features. In order to accentuate any sharp features, we fit the unconstrained design. The resulting difference estimator $\beta_i = d_i = y_{i+1} - y_i$, $i = 1, \dots, n - 1$ is best suited for identifying jumps - the sharpest possible data feature. Under the assumptions of independence and normality of ϵ

$$d_i = y_{i+1} - y_i \stackrel{iid}{\sim} N(\delta_i, 2\sigma^2)$$

where $\delta_i = E(d_i) = \theta_{i+1} - \theta_i$ for $i = 1, \dots, n - 1$.

Consider the fits produced with this design applied to noisy data from the smooth function and the piecewise constant function with jumps presented in Figure 1. The amount of zero-order signal captured in these fits is shown in economy plots of $|d_{(i)}|$, the index plot of the ordered first differences. This plot is related to the canonical BE plots through an orthogonal rotation. The economy plot for the piecewise constant function shows a strong signal for about a dozen large zero-order components, corresponding to the function's 11 jumps. Contrast this with the smoothly varying economy plot of the smoothly varying sinusoidal wave. The only apparent feature of this plot is its gradual slope corresponding to the contaminating noise.

We separate signal and noise estimated from this zero-order fit by controlling the False Detection Rate (FDR) for non-zero jumps (Benjamini and Hochberg 1995, Storey 2003). Zero-order knots are located between each data-pair judged to have a nonzero difference. This knot location problem is equivalent to the testing problem $H_O : \delta_i = 0, i = 1, \dots, n - 1$. The test statistics under the null hypothesis are $T_i = |d_i|/\sqrt{2\hat{\sigma}}$.

FDR adapts to sparse signals by controlling the expected proportion of spuriously included zero-order knots. When a data set is believed to have many sharp features we should not be overly concerned about the occasional inclusion of unnecessary knots, which is bound to happen. Instead it is preferable to control the number of falsely selected knots when there are few features detected.

Zero-order knots selected by FDR thresholding are added to a relatively coarse grid of linear or quadratic knots located at equally spaced sample quantiles. The i th row of the design matrix is

$$X_i = [1 \quad x_i \quad I(x_i > \omega_1) \cdots I(x_i > \omega_m) \quad (x_i - \kappa_1)_+^p \cdots (x_i - \kappa_k)_+^p], \quad (10)$$

where $p = 1$ or 2 , ω 's are the j zero-order knots, and the κ 's are the k 1-order knots $i = 1, \dots, n$. Assuming that most of the non-smooth features of the regression function have been identified with the zero-order knots, a relatively coarse grid of linear or quadratic order knots is often sufficient for

estimating smooth regions of the regression function. This model is fit with penalized least squares. This provides an additional opportunity for the penalty to “shrink” any spuriously selected zero-order knots. Thus, the penalty serves as a check against liberal choices of the nominal false detection rate, q . The smoothing parameter is selected by minimizing the estimated risk. The D_1 penalty matrix works well in practice.

AMPS is computationally efficient, requiring little more time to compute than traditional penalized regression splines. The number of operations necessary to locate the zero-order knots is relatively small and grow linearly with n .

3.2 Theoretical Support

Viewed as a penalized regression spline, the final multi-order design matrix of AMPS yields an admissible estimator (Cohen, 1996). Asymptotic results show that the adaptive first stage of AMPS is asymptotically minimax over a wide class of sparse normal mean problems, provided that the basis economy is asymptotically maximized.

For $0 \leq k \leq n$ let $e(k)$ denote the $n \times 1$ vector whose i th component is 1 if $1 \leq i \leq k$ and is 0 otherwise. The nested selection class $F_{NS} = \bigcup_{k=0}^n \{e(k)\}$ defines the set of shrinkage vectors for thresholding problems with a given basis, including the first stage AMPS fit. The canonical thresholding estimator is $\hat{\eta} = \hat{f}_{NS}z$. Beran (2000) describes the effect of basis economy on this estimator when $\hat{f}_{NS}$ is selected by minimizing estimated risk.

For every $b \in [0, 1]$, $r > 0$, and $\sigma^2 > 0$ consider the ball of smooth functions

$$B(r, b, \sigma^2) = [\eta : \text{ave}(\eta^2) \leq \sigma^2 r \quad \text{and} \quad \eta_i = 0 \quad \text{for} \quad i \geq bp].$$

A basis is economical if in the resulting canonical fit $\eta \in B(r, b, \sigma^2)$ for some value of b and $r > 0$.

Theorem 3.1 *Suppose that $\hat{\sigma}^2$ is consistent, $r > 0$, $\sigma^2 > 0$, and $b \in [0, 1]$. Denote the quadratic*

risk of estimator $\hat{\eta}$ as $R(\hat{\eta}, \eta, \sigma^2) = E|\hat{\eta} - \eta|^2$. Then, if $b \rightarrow 0$ as $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \frac{\sup_{\eta \in B(r, b, \sigma^2)} R(\hat{\eta}_{NS}, \eta, \sigma^2)}{\inf_{\hat{\eta}} \sup_{\eta \in B(r, b, \sigma^2)} R(\hat{\eta}, \eta, \sigma^2)} = 1. \quad (11)$$

The nested selection estimator $\hat{\eta}_{NS}$ achieves the lower bound on asymptotic minimax risk as basis economy is maximized. The proof follows directly from Theorem 4 in Beran (2000). Again, assume that $\hat{\sigma}^2$ is consistent, $r > 0$, $\sigma^2 > 0$, and $b \in [0, 1]$. For fixed b , the lower on the asymptotic minimax risk for $\hat{\eta}_{NS}$ is

$$\lim_{n \rightarrow \infty} \sup_{\eta \in B(r, b, \sigma^2)} R(\hat{\eta}_{NS}, \eta, \sigma^2) = \sigma^2 \min(r, b).$$

The asymptotic minimax risk over all estimators is

$$\lim_{n \rightarrow \infty} \inf_{\hat{\eta}} \sup_{\eta \in B(r, b, \sigma^2)} R(\hat{\eta}, \eta, \sigma^2) = br\sigma^2/(r + b).$$

In related work, Abramovich et al. (2002) prove that FDR driven threshold estimators are asymptotic minimax when the false detection rate q asymptotically approaches zero.

The adaptation of the first stage thresholding can be thought of as an initial zero-order basis economization step. Assuming that the unknown regression function has smooth regions, we add smooth basis functions to refine the zero-order basis. While this strategy performs well in practice, we have not formally established its asymptotic properties. Considered as a single hybrid shrinkage estimator AMPS applies hard thresholding to select zero-order components and a monotone shrinkage class is used to estimate the smoother second state components. Thus, AMPS is related to Effromovich's hybridized wavelets (1999) and the hybrid shrinkage estimators of Beran (2003), both asymptotically minimax estimators. We conjecture that AMPS inherits similar asymptotic properties.

4 Variance Calibration

Variance Calibration (VC) is a Monte Carlo technique for adaptively excluding local biases associated with poorly fit regression features or outliers. The VC estimator $\tilde{\sigma}^2$ is a robust variance

estimator particularly well suited for spatially heterogenous regression functions.

Consider the class of linear estimators $\hat{\theta} = \hat{m}(x) = Sy$ with smoothing matrix S . The estimator of σ^2 most commonly associated with S is a function of the squared residuals divided by the degrees of freedom of the fit,

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (Y_i - \hat{m}(x_i))^2}{n - df(S)}, \quad (12)$$

where $df(S)$ is the degrees of freedom of the smoother S . For the least squares estimator, the degrees of freedom correspond to the rank of the estimator, S . Buja, Hastie and Tibshirani (1989) discuss other definitions for the degrees of freedom associated with the linear estimator, S .

When the linear estimator S fits well, the residuals are good estimates of the noise in the data, and $\hat{\sigma}^2$ is expected to perform well. However, when the smoothing basis is not flexible enough, regions of the data will be oversmoothed, producing residuals that are contaminated by the signal. Our goal is to exclude these signal-contaminated residuals.

Since the sharpest regression features create the largest biases in the estimated variance, we focus our attention on identifying these features. First differences are a convenient basis for estimating sharp features. Indeed, first differences can be thought of as the residuals from a severely undersmoothed fit. Formally, we define the first differences as $d_i = y_{i+1} - y_i$, $i = 1, \dots, n - 1$ with distribution $d_i \sim N(\delta_i = \theta_{i+1} - \theta_i, 2\sigma^2)$.

Consider the estimation of the differences δ_i from d_i where there are two distinct components: noise components and signal-contaminated components. Noise components have zero expectation, $E(d_{(i)}) = \delta_{(i)} = 0$ where $i = 1, \dots, \lfloor \psi \rfloor$, ψ is the signal-noise threshold, and $d_{(i)}$ are the first differences ordered from smallest to largest in absolute value for $i = 1, \dots, n - 1$. This is the set of first differences uncontaminated by signal; these differences correspond to the smoothest regions of the regression surface. The remaining first differences have strictly positive absolute expectation, $E(|d_{(i)}|) = |\delta_{(i)}| > 0$ for $i = \psi + 1, \dots, n - 1$. This estimation problem is referred to as a thresholding problem or a subset selection problem (Hastie et al. 2001). We will refer to it as the sparse normal

mean problem (Johnstone 1994).

If ψ is known we can estimate σ^2 with the first difference variance estimator, treated in Rice (1984),

$$\hat{\sigma}^2(d|c) = \frac{1}{2c} \sum_{i=1}^c d_{(i)}^2. \quad (13)$$

Under the sparse normal mean problem this variance estimator is consistent provided that $c \leq \psi$. Conversely, if σ^2 is known ψ may be estimated as the largest cutoff c such that $\hat{\sigma}^2(d|c) \leq \sigma^2$ in (11).

Because σ^2 and ψ are seldom known in practice our strategy is to estimate ψ by construcing a related problem where variance is known. This estimate, $\tilde{\psi}$, is then used to estimate the unknown variance σ^2 . In what follows we introduce the VC estimator and prove its consistency.

Letting $y_i^* = y_i + \alpha_i$ be the Monte Carlo augmented data where $\alpha \stackrel{iid}{\sim} N(0, \sigma_{VC}^2)$ for a specified value of σ_{VC}^2 . Define the augmented differences to be $d_i^* = y_{i+1}^* - y_i^*$ so that

$$d_i^* \stackrel{iid}{\sim} N(\delta_i = \theta_{i+1} - \theta_i, 2(\sigma^2 + \sigma_{VC}^2)).$$

Let $d_{(j)}^*$ denote the augmented differences ordered from smallest to largest in absolute value for $j = 1, \dots, n-1$. Note that both sets of differences share the same signal-noise threshold, ψ .

Estimate the signal-noise threshold, $\tilde{\psi}$, as the largest cutoff c for which the difference in variances of differences (DVD)

$$\begin{aligned} DVD(c) &= \hat{\sigma}^2(d^*|c) - \hat{\sigma}^2(d|c) \\ &= \frac{1}{c} \sum_{j=1}^c d_{(j)}^{*2} - \frac{1}{c} \sum_{i=1}^c d_{(i)}^2 \end{aligned}$$

is less than or equal to σ_{VC}^2 . Different indices are used to emphasize the fact that each set of differences is independently ordered. Therefore, the variance calibration algotihm is

1. Calculate $d_i = y_{i+1} - y_i$,

2. Generate the Monte Carlo augmented data $y_i^* = y_i + a_i$ where $a \stackrel{iid}{\sim} N(0, \sigma_{VC}^2)$,
3. Let c_j for iteration j be the smallest cutoff c such that $\hat{\sigma}^2(d^*|c) - \hat{\sigma}^2(d|c) \geq \sigma_{VC}^2$.

The Monte Carlo variation behind the estimator is controlled by iterating the preceding steps over m independent realizations and estimating $\hat{\psi}$ as the expected cutoff. Calculating several Monte Carlo estimates has the added benefit of allowing the user to quantify the Monte Carlo variance of the procedure.

- 4 Repeat Steps 2 – 3 m times and record the estimated cutoff, c_i for $i = 1, \dots, m$.
- 5 Compute $\hat{\psi} = \lfloor \text{mean}(c_i) \rfloor$.

The quantity σ_{VC}^2 needs to be specified large enough to be estimable while not being so large as to “wash out” the signal in d_i . Empirical studies show that the structure of d_i is preserved with a choice of $\sigma_{VC}^2 \approx \sigma^2$. Although this specification is unlikely, it suggests that a simple, initial estimate of variance may be used for σ_{VC}^2 . Although simple estimates of σ^2 will often be positively biased in the presence of sharp features or outliers, simulations show that the VC procedure is much less sensitive to very large values of σ_{VC}^2 , relative to σ^2 , than it is to very small values (Fink 2003).

4.1 Properties of VC Estimator

In what follows we prove that both the DVD and VC estimators are unbiased and consistent. We briefly discuss how these results apply to real data that may not meet the assumptions of the sparse normal mean problem.

The expected behaviour of the DVD estimator is derived by matching the moments of the first differences. Scaled to unit variance, both first differences share the same distribution

$$\frac{d_i^*}{\sqrt{2(\sigma^2 + \sigma_{VC}^2)}} \stackrel{D}{=} \frac{d_i}{\sqrt{2\sigma^2}} \stackrel{iid}{\sim} N(\delta_i, 1).$$

Consequently, the scaled order statistics order statistics also share the same distribution,

$$\frac{d_{(i)}^2}{2(\sigma^2 + \sigma_{VC}^2)} \stackrel{D}{=} \frac{d_{(i)}^{*2}}{2\sigma^2}. \quad (14)$$

Taking expectations of Equation (11) above yields the following equality

$$2\sigma^2 E(d_{(j)}^{*2}) = 2(\sigma^2 + \sigma_{VC}^2) E(d_{(i)}^2).$$

Rearranging, so that the difference between expectations is on the left hand side,

$$E(d_{(j)}^{*2}) - E(d_{(i)}^2) = \sigma_{VC}^2 \frac{E(d_{(i)}^2)}{\sigma^2}. \quad (15)$$

Under the canonical model, the above simplifies to

$$E(d_{(i)}^{*2}) - E(d_{(i)}^2) = \begin{cases} 2\sigma_{VC}^2 & \text{if } c \leq \psi \\ \sigma_{VC}^2 \left(2 + \frac{\delta_i^2}{\sigma^2}\right) & \text{if } \psi + 1 \leq c \leq n - 1. \end{cases} \quad (16)$$

Summing over each ordered set yields

$$E \left[\frac{1}{c} \sum_{i=1}^c d_{(i)}^{*2} - \frac{1}{c} \sum_{i=1}^c d_{(i)}^2 \right] = \begin{cases} 2\sigma_{VC}^2 & \text{if } c \leq \psi \\ \sigma_{VC}^2 \left(2 + \frac{\sum_{i=1}^c \delta_i^2}{c\sigma^2}\right) & \text{if } \psi + 1 \leq c \leq n - 1. \end{cases}$$

Consequently, $DVD(c)$ is consistent provided $c \leq \psi$ or that c is small enough such that $\sum \delta_i^2/c \rightarrow 0$ as n tends to infinity.

Assuming that each set of ordered differences is independent, the law of large numbers insures that the $DVD(c)$ estimator converges in probability,

$$DVD(c) = \frac{1}{c} \sum_{j=1}^c d_{(j)}^{*2} - \frac{1}{c} \sum_{i=1}^c d_{(i)}^2 \rightarrow \frac{1}{c} \sum_{i=1}^c E(d_{(i)}^{*2}) - \frac{1}{c} \sum_{i=1}^c E(d_{(i)}^2),$$

as $n \rightarrow \infty$. Although the two sets of differences are, by construction, independent of each other, they are serially correlated among themselves due to the ordering of the sequences. The effect of this ordering on the correlation diminishes as c approaches n , meaning that an assumption of independence will often be achieved simply by assuming that c or ψ is large relative to the sample size n .

Additionally, we assume that data density increases uniformly across the domain such that $\psi/n \rightarrow \psi^*$, as n tends to infinity and ψ^* is a constant on $[0, 1]$. Then $\hat{\psi}$ will be consistent for ψ^* and the VC estimator $\tilde{\sigma}^2 = \hat{\sigma}^2(d|\hat{\psi})$ is consistent for σ^2 .

The difference between real data and canonical thresholding data is that there may not be a clear cut way to divide signal from noise. There may be some bias contamination for all components. As expected, the performance of the VC estimator degrades as a function of the degree of signal contamination.

5 Simulations

Simulations were carried out to examine the performance of the AMPS and compare it to several other methods in the literature. For all simulations it is assumed that $Y \sim N_n(m(x), \sigma^2)$ with both $m(x)$ and σ^2 unknown. The first set of simulations fits smooth functions punctuated by sharp discontinuities and relatively large noise levels. The second simulation test compares AMPS performance with other adaptive methods on the Wavelet test function suite of Donoho and Johnstone (1995). This suite of spatially heterogeneous functions have relatively high signal to noise ratios. In the third simulation, AMPS performance is tested on a set of spatially homogeneous functions. In the final example we demonstrate the performance of AMPS with a two-way design.

All AMPS fits are calculated using FDR $q = 0.05$ and $\hat{\sigma}_{VC}^2$ is computed with an initial estimate from a linear PRS. All simulations are computed in Matlab 5.3 (Mathworks 2001).

5.1 Smooth Functions with Jump Discontinuities

The first set of simulations consists of four smooth functions punctuated by jump discontinuities with relatively low signal-to-noise ratios. Signal-to-noise ratios are measured as $sd(m(x))/\sigma$, where $sd(m(x))$ is the estimated standard deviation of the regression function, $m(x_i)$ over the sample $i = 1, \dots, n$ and σ is the true standard deviation of the noise in the data.

Table 1: Specifications of Smooth functions with Jump Discontinuities

$m(x)$	x	σ^2	$sd(m)/\sigma$	$h/2\sigma$
$m(x) = \begin{cases} 0 & 0 < x < 0.3 \\ -1 & 0.3 < x < .7 \\ 0 & \text{o.w.} \end{cases}$	$[0,1]$	0.12	4.0907	4.1667
$m(x) = \begin{cases} 1.75x^2 & x < 0.5 \\ .25x + & \\ 0.025\sin(10x^4) & \text{o.w.} \end{cases}$	$[0,1]$	0.0296	3.1667	4.7279
$m(x) = X_1 b_1$ where	$[0,5]$	1.0890	3.500	5.2338
$X1_i = [I(x_i < 5) \ I(x_i \geq 5) \ x_i^2 I(x_i < 5) \\ (x_i - 2)^3 I(x_i < 5) \ x_i I(x_i > 6) \\ x_i^5 I(x_i > 7) \ x_i I(x_i > 7.5)]$				
$b_1 = [-5 \ -10.5 \ -.51 \ -41.5]$				
$m(x) = \begin{cases} -18x^2 + 5 & x < 0.25 \\ 1.25 & 0.25 < x < 0.7 \\ 15(x - 1)^2 & \text{o.w.} \end{cases}$	$[0,1]$	0.3112	3.8333	3.0549

The true regression functions, domains, noise levels, and signal-to-noise ratios are presented in Table 5.1. For this set of simulations $n = 250$ and the x 's are equally spaced over the indicated domains. Errors, ϵ_i are independent $N(0, \sigma^2)$. The number of second stage knots is $k = 50$. For comparison, we also fit the spatially adaptive penalized regression splines (SA-PRS) of Ruppert and Carroll (2000). The number of knots is set to 50 with 13 subknots placed at equally spaced quantiles of the data. The number of iterations $N_{iter} = 1$ and quadratic splines are used.

We simulate 250 data sets and for each replication we calculate the two spline estimates of the function and the associated mean squared error (MSE),

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{m}(x_i) - m(x_i))^2.$$

Typical realizations of the data, fits, and boxplots for each of the four functions are shown in Figure 2.

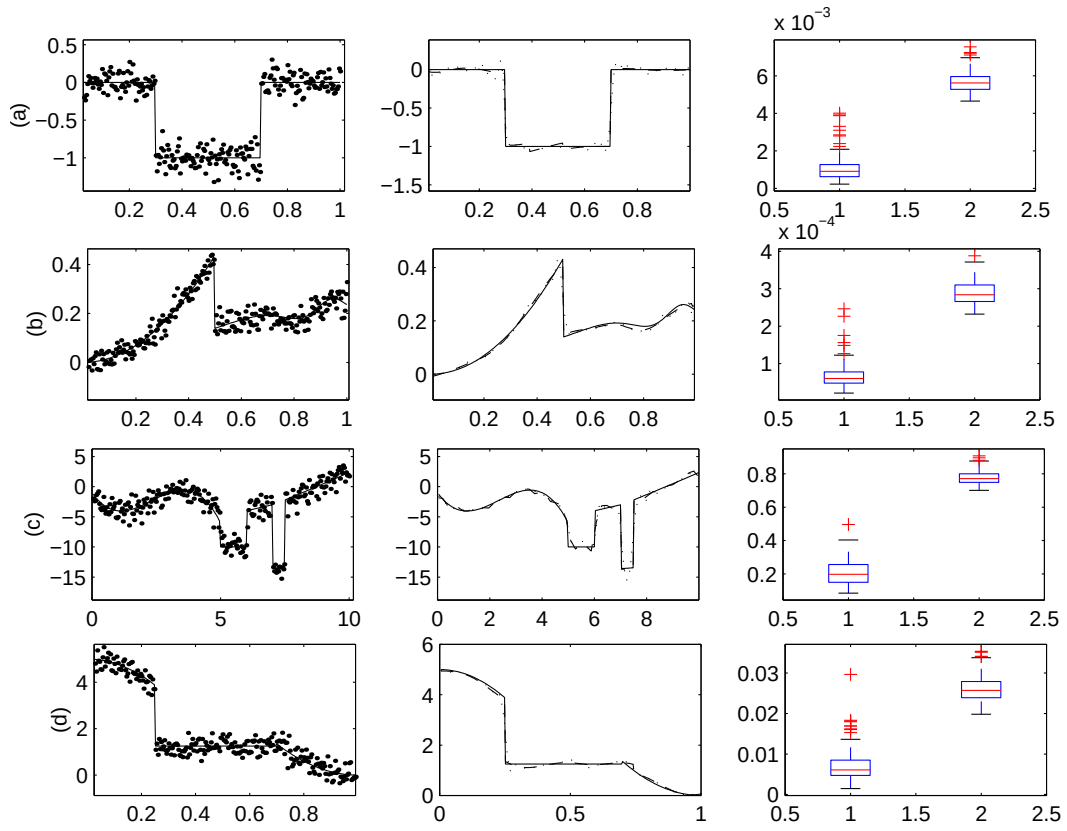


Figure 2: Smooth Functions with Jump Discontinuities. Each of four functions (a)-(d). Columns from left to right: true function with noisy data, true function(solid) with AMPS (dashed) and SA-PRS (dotted), box plots for AMPS (left) and SA-PRS (right).

We see that AMPS does identify the jumps in these functions and AMPS outperforms SA-PRS in the presence of jumps. This performance difference depends on the number and relative size of the jumps.

If the height of a jump discontinuity is h then we define the jump ratio as the ratio of the height of the jump to twice the standard deviation of the noise in the vicinity of the jump, $h/(2\sigma)$. The jump ratio provides a measure of the size of a jump relative to the amount of noise in the data. Thus, the jump ratio should tell us something about the ability to detect a jump or spike in the data. It would be difficult for *any* method to detect jumps with a jump ratio of two or less. The same is true for AMPS. As the jump ratio increases AMPS performance improves as a function of FDR performance. For example, a simulation of a step function with a jump of height $h = 0.5$, $n = 100$ for x 's equally spaced over $[0, 1]$, $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$, and σ^2 scaled such that the jump ratios tested are 1, 2, 3, 4, and 5 was run for 100 repetition for each jump ratio. The jump at 0.5 was detected for 11, 33, 82, 100, and 100 percent of the trials for each jump ratio, respectively.

5.2 The Wavelet Function Suite

The second battery of simulations are based on the Wavelet test functions of Donoho and Johnstone (1995): Blocks, Bumps, HeaviSine, and Doppler. The strong spatial heterogeneity of these functions have made them an attractive suite of test functions for many recent smoothing methodologies. We analyze these functions to examine the performance of the AMPS and compare several other methods in the literature. Separate experiments are performed for small and large sample sizes.

The first simulation compares the performance of AMPS and SUREshrink Wavelets of Donoho and Johnstone (1995) with small sample sizes. Sample sizes of $n = 128$ and $n = 256$ are used with equally spaced x 's over $[0, 1]$. The test functions are scaled so that the signal-to-noise ratio is approximately 7, as in experiments 1 through 4 from Luo and Wahba (1997) with independent errors $N(0, \sigma^2 = 1)$.

The AMPS fit is computed with $k = \lfloor n/2 \rfloor$ linear knots. The large number of second stage knots are used in order to fit the highly variable underlying functions. Root mean squared errors (RMSE) for 20 replications are computed. In Table 2 we summarize these results and report the smallest and largest RMSE from the corresponding experiments from Donoho and Johnstone (1995). A typical realization of the Wavelet test functions for $n = 250$ with equally spaced x 's over $[0, 1]$, and AMPS fits for $k = 40$ are shown in Figure 3. The AMPS (dashed) fit uses 40 linear knots with a false detection rate of 0.05. Zero-order knot locations are indicated by the triangles on the x -axis.

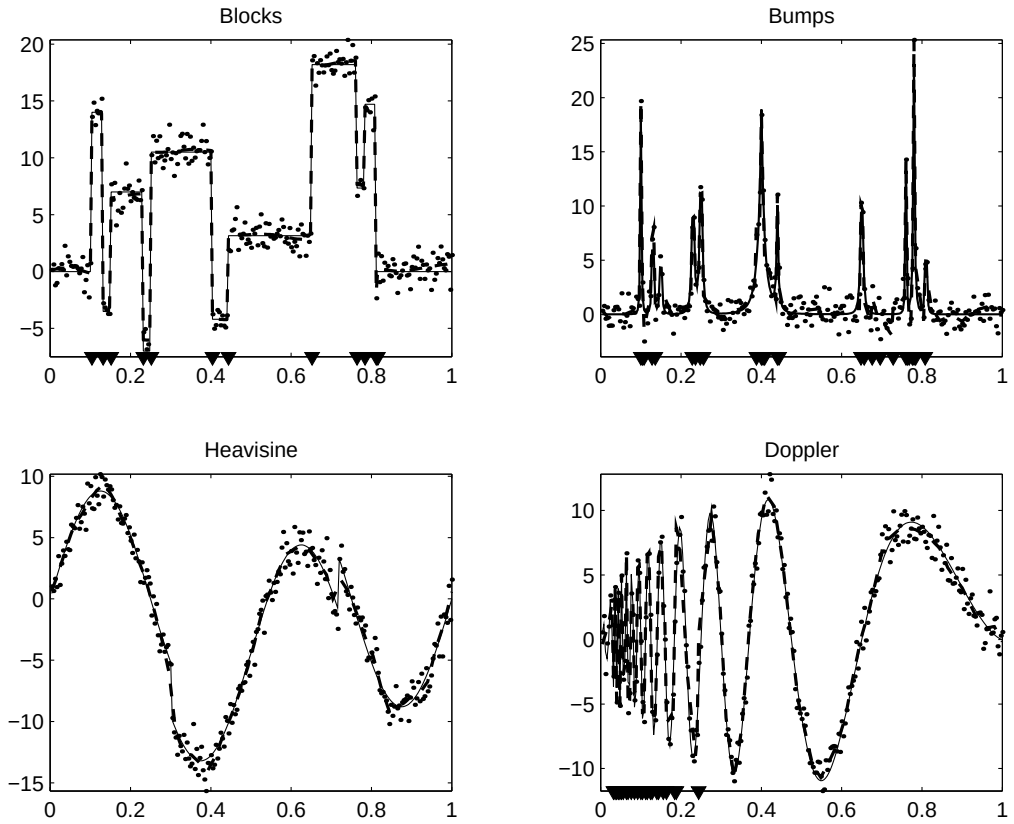


Figure 3: The AMPS(dashed) fit uses 40 linear knots with a false detection rate of 0.05. Zero-order knot locations are indicated by the triangles on the x -axis.

We perform experiments 1-4 the same Luo and Wahba (1997) to demonstrate the AMPS performance. Thirty-one replicates are performed with each of the wavelet functions with $n = 2048$.

Table 2: Root Mean Squared Errors of Estimation for the Wavelet test functions with $n = 128$ and $n = 256$.

$m(x)$	n	$DJ - Small$	$DJ - Large$	$AMPS$
Blocks	128	.81	1.18	.4592
	256	.68	1.20	.3819
Bumps	128	.93	1.35	.9289
	256	.85	1.41	.8414
HeaviSine	128	.61	.77	.5345
	256	.49	.62	.4355
Doppler	128	.80	.94	1.0857
	256	.74	.98	.9521

AMPS is fit to these data sets with $k = \lfloor n/2 \rfloor$. MSE's are reported in Table 5.3 along with results from several other adaptive methods. From the results in Tables 5.2 and 5.3 we see that

- AMPS performs well compared to SUREshrink with small sample sizes,
- AMPS performs well compared to SUREshrink for Blocks across all sample sizes,
- AMPS performance degrades, relative to SUREshrink, with increasing sample size for other functions. This is especially true for the Bumps and Doppler functions where the fine structure of the functions are themselves revealed with larger sample sizes.

The degradation in relative performance of AMPS for the wavelet functions point out a fundamental limitation of AMPS. The AMPS's piecewise polynomial basis is best suited to fitting jumps, spikes, and smooth regions. The spatial heterogeneity exhibited by Bumps and Doppler is smoother than the simple jumps of Blocks. Moreover, the fine details of these functions is revealed only with larger sample sizes. This accounts for the difference in relative performance between AMPS and SUREshrink across different sample sizes. AMPS consistently locates and approximates the fine

Table 3: Median of mean squared errors and the difference of the first and third quartiles of MSE (in parentheses) for the Wavelet test functions with $n = 2024$. Methods used include Hybrid Adaptive Splines (HAS) of Luo and Wahba (1997), cubic smoothing splines (SS), SUREshrink, and the Automatic Bayesian Splines (ABS) of Denmsion, Mallick and Smith (1998).

Experiment	HAS	SS	SUREshrink	ABS	AMPS
Blocks	.137 (.018)	.546 (.023)	.398 (.049)	.170	.0390 (.0077)
Bumps	.087 (.021)	.124 (.010)	.167 (.015)	.167	.5599 (.0773)
HeaviSine	.039 (.013)	.075 (.005)	.062 (.007)	.033	.1007 (.0296)
Doppler	.068 (.015)	.205 (.011)	.145 (.013)	.135	.3325 (.0166)

structure of these functions, but as the sample size increases, it becomes more apparent that the AMPS basis can not fit these functions as smoothly as the wavelet bases of SUREshrink.

There are several possible fixes for this deficiency in AMPS. The simplest is to increase the number of knots used during the second stage fit. This gives the final regression spline model more flexibility in those regions where the functions have the greatest spatial heterogeneity.

5.3 Smooth Functions

The third simulation tests the performance of AMPS when the underlying function is smooth. To compare AMPS with other methods we considered Case 6 in Luo & Wahba (1997). The regression function there is from Fan and Gijbels (1995),

$$m(x) = \sin(2(4x - 2)) + 2\exp(-16^2(x - 0.5)^2).$$

We use $\sigma = .3$ and $n = 256$ with equally spaced x 's on $[0, 1]$. The results for 400 replications are presented in Table 5.4 along with the results from Luo & Wahba's (1997) hybrid adaptive spline, smoothing splines (SS), SUREshrink of Donoho and Johnstone (1995), MARS of Friedman (1991), and the spatially adaptive penalized regression splines (SA-PRS) of Ruppert and Carroll (2000).

Table 4: Median of mean squared errors and the difference of the first and third quartiles of MSE (in parentheses) of the Smooth test function.

HAS	SS	SUREshrink	MARS	Local PRS	Global PRS	AMPS
.007	.006	.018	.007	.0053	.0061	.0067
(.006)	(.003)	(.004)	(.004)	(.0035)	(.0029)	(.0034)

Denison, Mallik and Smith (1998) tested their Automatic Bayesian splines on the same example except they used $n = 200$ and they reported the average MSE. They found average MSE values of 0.0096 and 0.0087 for linear and quadratic splines, while our average MSE value was .0105 with standard deviation .0168 for this experiment.

5.4 A 2-dimensional AMPS Extension

The two dimensional AMPS estimator is a direct extension of the one dimensional procedure presented in Section 3. Zero-order knots are simply added to a linear tensor product regression spline and fit with penalized least squares.

The set of basis functions for the one dimensional p th order regression spline is

$$\mathcal{B} = [1, x_i, \dots, x_i^p, (x_i - \kappa_1)_+^p, \dots, (x_i - \kappa_k)_+^p].$$

Let $\mathcal{B}(i)$ denote the corresponding set of basis functions for dimension i . The tensor-product regression spline basis is defined to be $\mathcal{B}(1, 2) = \mathcal{B}(1) \otimes \mathcal{B}(2)$, the set of all products $b(1) \dots b(j)$ where $b(j) \in \mathcal{B}(j)$. Thus, the two-dimensional tensor-product PRS model can be written as

$$\begin{aligned} f(x, z|\beta) = & \beta_0 + \sum_{i=1}^p \beta_{1,i} x^i + \sum_{i=1}^p \beta_{2,i} z^i + \\ & \sum_{i=1}^k \beta_{3,i} (x - \kappa_i)_+^p + \sum_{i=1}^k \beta_{4,i} (z - \kappa_i)_+^p + \sum_{i=1}^k \sum_{j=1}^k \beta_{5,ij} (x - \kappa_i)_+^p (z - \kappa_j)_+^p, \end{aligned}$$

where for convenience we use the same set of knots κ for both dimensions.

Similar to the one dimensional method, all truncated spline bases are penalized. Delaunay triangulation is used to define neighbors between which the first differences are computed. These

first differences are used to locate zero-order knots and estimate σ^2 with VC calibration. This estimator works for evenly and unevenly spaced data. Similar to the one-way layout, this estimator is particularly well suited for the estimation of smooth surfaces punctuated by sharp features like spikes or edges.

We analyze the regression function

$$m(x, z) = \begin{cases} 5 + \frac{8}{3} \sin(4z) \cos(16x) & x, z \in [0.5, 1]^2 \\ 0 & \text{otherwise.} \end{cases}$$

with $\sigma = .8$ and $n = 30$ with equally spaced x_i and y_i on $[0, 1]^2$, $i = 1, \dots, n$ producing a sample size of 900. We use a tensor product two-way tensor product spline basis based on an equally spaced grid on $[0, 1]^2$ of 225 knots. The true regression surface, interpolated data and the AMPS fit are shown in Figure 4.

6 Extensions

AMPS can be used to estimate the location and number of changepoints. As a nonparametric changepoint estimator, AMPS is an attractive tool because it controls the false detection rate for changepoints without imposing strong restrictions on the regression relationships between the changepoints. Inferences about the changepoints themselves may be carried with bootstrap analysis over the selected zero-order knots. For example, Fink and Hairston (2004) use AMPS to investigate the effects of invasive zooplankton species on the timing of the clear water phase in Onondaga Lake, New York. In ongoing work, AMPS is used to identify differentially expressed genes from cDNA microarray data.

By emphasizing the connections between penalized regression splines and linear mixed models Ruppert et al. (2003) are able to greatly extend the usefulness of the PRS model. As a simple extension of the PRS, AMPS can also be extended to include penalized and unpenalized covariates. Future work will investigate extending AMPS for nonnormal responses. AMPS can be extended to the heteroskedastic sparse normal mean problem simply by using weighted penalized least squares

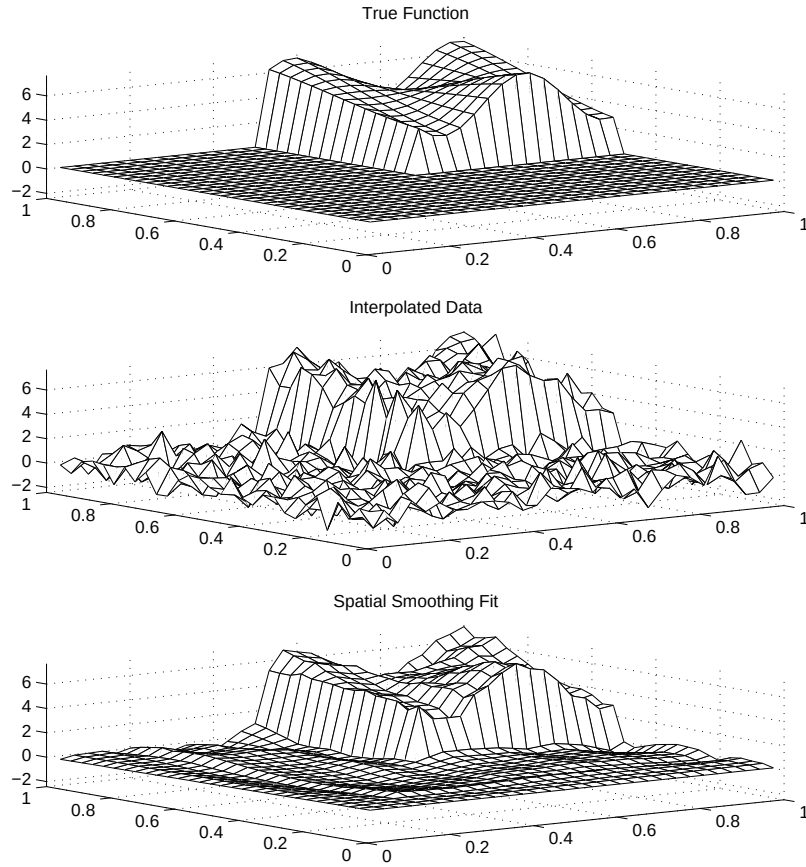


Figure 4: The true regression surface, interpolated data and the AMPS fit.

fit with iteratively reweighted least squares. Utilizing the Bayesian interpretation of the PRS, we can compute pointwise confidence intervals that take into account the uncertainty in estimating both the smoothing parameter as well as the location zero-order knots (Ruppert and Carroll 1997 and Fink 2003).

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